

Network Storage with Raspberry Pi

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An always-on network backup device accessible to a computer from both outside and inside the network is a highly desirable thing. As the system will always be on, power consumption is an important consideration. These requirements are fairly met by this Raspberry Pi-based network storage system as it consumes less power than its off-the-shelf counterparts.

Presented here is a system which uses two USB hard drives plugged into Raspberry Pi to store data. One of the hard drive stores data directly and the other is used to back up the data.

For a simple network, one drive

is sufficient but it is recommended to use two for local redundancy. External hard drives should be low-power and have sufficient capacity for future addition of data. In this case, two NTFS-formatted hard drives are used and Samba software helps to implement network sharing. Follow the steps mentioned below in the exact sequence to set up and configure the network storage system with Raspberry Pi.

Mounting the external hard disk

Connect the hard drives to the Raspberry Pi and power it up. It is a good idea to use a USB hub with separate power supply. The Raspberry Pi

should have a working operating system (Raspbian) configured suitably for remote access to follow all the steps. You can refer 'Remote Access to Raspberry Pi' published in June 2013 issue.

Now access Raspberry Pi using SSH and run following commands to add the support for NTFS-formatted drives to Raspbian operating system as shown in Fig. 1:

```
sudo apt-get
install ntfs-3g
```

It will take a few minutes for the package to download, unpack and install. Once the NTFS package is installed, look for the un-mounted partitions of the attached external hard drives



by executing the following command as shown in Fig. 2:

```
sudo fdisk -l
```

If only one external hard drive is connected, you will see two disks—one external hard disk and one SD card inside Raspberry Pi. In case two external hard drives are connected, you will see three disks. Carefully make note of the hard drives' names.

Before mounting the drives, we need to create a directory to mount the drives to. For the sake of simplicity, let us create a directory called HDD1 and HDD2 for each drive. To create these directories, write the commands mentioned below and shown in Fig. 3:

```
sudo mkdir /media/HDD1
```

```
sudo mkdir /media/HDD2
```

You can mount the drives to each location by executing the following commands:

```
sudo mount -t auto /dev/sda1
```

```
/media/HDD1
```

```
sudo mount -t auto /dev/sdb1
```

```
/media/HDD2
```

'sda1' and 'sdb1' are names of the external drives; replace them with the ones you get as shown in Fig. 2.

Now we have both the drives mounted; we just need to create a folder in both directories to hold all the shared folders. Refer Fig. 4 and write the commands:

```
sudo mkdir /media/HDD1/shares
```

```
sudo mkdir /media/HDD2/shares
```

Setting up the network

Install Samba for accessing the storage from anywhere on the network using the command below as shown in Fig. 5:

```
sudo apt-get install samba
samba-common-bin
```



Fig. 1: Add NTFS support

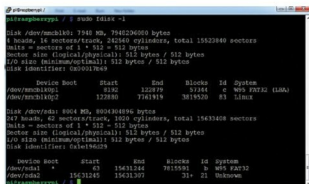


Fig. 2: Partition of the external hard drives



Fig. 3: Make the directories to mount the drives

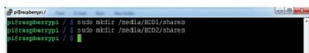


Fig. 4: Create the shared folder



Fig. 5: Installing Samba